Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/02/2025
PI Name:	Blackwell, Ashley Ph.D.		
Project Title:	Feasibility Study: Use of Neural Networks to Predict Adaptability and Multiday Performance Saving in Dual Motor-Cognitive Tasks After Exposure to Space Flight Stressors		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Neurobiology		
Space Biology Special Category:	None		
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Zip Code:	89154	**Congressional District:**	1
Comments:	PI moved to University of Nevada, Las Vegas		
Project Type:	Ground	**Solicitation / Funding Source:**	2021 Space Biology NNH21ZDA001N-SBAS E.11: Animal Studies
Start Date:	08/01/2023	**End Date:**	12/01/2025
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Jones, Harry	**Contact Phone:**	650.604.5518
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/01/2025 per F. Hernandez/ARC (Ed., 9/17/25). NOTE: End date changed to 03/31/2025 per NSSC information (Ed., 6/13/24).		
Key Personnel Changes/Previous PI:	No CoI This Year		
COI Name (Institution):			
Grant/Contract No.:	80NSSC23K1314		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This proposal will use a rodent model of space flight stressors (SFS), including space radiation (SR) and sleep disruptions (SD), which are known to impair mission-relevant performance that is dependent on cognitive and sensorimotor systems. Astronauts must engage in complex tasks that rely on the integration of information from these multiple systems simultaneously, both independently and cooperatively in team cohesion, while on deep space missions. Engagement in multiple tasks at once, or dual tasks, are very common, such as navigating (i.e., walking or floating in space) through the environment while operating a device (i.e., tablet or radio). Independent, or individual, dual task performance has been shown to be disrupted in astronauts on both short and long duration space missions that was attributed to microgravity. Yet it is not known how SR and SD will impact performance during complex tasks, including adaptations to new stimuli and re-adaptations to old stimuli, nor whether these deficits will extend to team cohesion beyond the individual, which is critical for performance and mission success. Surprisingly, no work to-date has examined dual task performance in a rodent model of SFS. Dual tasks involve performance in multiple systems simultaneously and provide the opportunity to evaluate adaptations, re-adaptations, and multiday performance savings in rodents and humans. Savings refers to faster relearning, or gains in performance, that come from repetition on a task. Both savings and adaptations are imperative to mission success with exposure to SFS and varying task demands. Therefore, the main objectives of this proposal are two-fold: 1) to evaluate the impact of SFS, alone and combined, on neural activity during dual tasks, periods of inactivity, and sleep, and 2) to assess the feasibility of using neural activity to predict subsequent adaptations and multiday performance savings. To accomplish these research objectives, neural recording techniques will be used to establish system-wide SFS effects on independent and cooperative (team cohesion) dual task performance and to characterize the neural mechanisms underlying adaptations and savings, as well as the feasibility of predicting future performance. Neural recording techniques will be conducted while rats perform versions of behavioral assessments that vary in complexity and during offline periods of inactivity (i.e., neural replay) to examine cognitive, sensorimotor, and vestibular function. Sleep disruptions are commonly reported among astronauts, which have deleterious effects on performance, including reaction time. Therefore, we will also investigate the initial effects of SR exposure on neural activity. In addition, we will assess the feasibility of predicting neural network function, adaptation, and savings, from baseline SR neural recording. Despite both independent and cooperative (team cohesion) performance depending on varying demands (single versus dual) in many of the tasks that astronauts must perform in space, the impact that SR and SD have on such performance is unknown. Thus, our proposed studies will determine the relative sensitivity of independent and cooperative (team cohesion) performance on single and dual tasks to these SFS, compared to mono-dimensional tasks that have been the mainstay of rodent-based research to-date. This work also has the potential to identify underlying neurobiological mechanisms of adaptations in a rodent SFS model and to provide a basis to identify resilient and susceptible factors. This work may lead to an understanding of how to promote adaptations and performance savings across time in individuals that are especially susceptible to the effects of SFS.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	No work to-date has evaluated dual motor-cognitive performance in rodent models, including those of space flight stressors (SFS), to determine the neurobiological basis for the disruptions caused by SFS or how they impact adaptations, savings, and the predictability thereof using neural network function. Given the widespread effects of SFS, understanding how adaptations, readaptations, and savings occur within multiple systems (sensorimotor, cognitive, social cooperative) in rats identified to be resilient or susceptible will be imperative to develop effective therapeutic strategies. This will be the first study to evaluate both independent and cooperative, team cohesion, performance in tasks of sensorimotor and cognitive function in a rodent model. Performing both autonomously and cooperatively with a team is critical for long duration missions in deep space along with determining how to maximize team dynamics. In addition, the proposed work will fill a large gap in the acute effects of SFS immediately after space radiation exposure by examining behavior and neural activity acutely after post-irradiation exposure out to protracted timepoints. The implications of this work will extend beyond the space biology animal studies program to sleep disruptions that commonly occur on Earth, neural network dysfunction, and adaptation, readaptations, and savings in patients on Earth. Specifically, many of the same impairments observed after space radiation exposure and sleep disruptions are also observed in rodent models of stroke; it is likely that a similar mechanistic basis exists for these deficits as well as the compensation in performance that occurs over time. Further, since this will be the first study to evaluate dual motor-cognitive tasks during independent and cooperative team work in a rodent model, it will provide validation for the use of these assessments in other rodent models outside of SFS. Therefore, this work will have applications for rodent models of stroke and clinical populations. Lastly, evaluating neural network function in rodent models of ground based SFS will provide a basis of comparison for functional magnetic resonance imaging studies with astronauts.
Task Progress:	Note: The Principal Investigator (PI) reports that delays were experienced in this fiscal year (ordering and receiving equipment, etc.), causing certain scheduling changes to this work. See the Cumulative Bibliography to download conference presentations related to this project (Ed., 4/8/25). All independent and cooperative post-radiation pre-screening behavioral testing has been completed, and rats are undergoing surgery, which will be followed by post-surgery behavioral assessments.
Bibliography Type:	Description: (Last Updated: 04/08/2025)
Conference Materials (Downloadable)	Rodgers DH, Attallah NM, Adhikari A, Garcia TL, Moges NA, Omengan KD, Cortez JF, Dhede KS, Blackwell AA. "Dual cognitive motor task performance in sleep fragmented sham and irradiated Wistar rats." Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 HRP IWS 2025_Abstract Rodgers, Doc-Haden.pdf
Conference Materials (Downloadable)	Garcia TL, Moges NA, Ordonez S, Attallah NM, Rodgers DH, Adhikari A, Omengan KD, Lloyd D, Gleed SL, Blackwell AA. "Two rats, one string: Cooperative behavior after exposure to space flight stressors." Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 HRP IWS 2025_Abstract Garcia, Teresa.pdf

Conference Materials (Downloadable)	Lloyd D, Cortez JF, Garcia TL, Adhikari A, Attallah NM, Moges NA, Rodgers DH, Dhede KS, Omengan KD, Blackwell AA. "Impact of radiation (4He) exposure and sleep fragmentation on memory and motor skill adaptation in rat string-pulling behavior." Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025 HRP IWS 2025_Abstract Lloyd, Diana.pdf